



Chitkara

PAPER OF JAVA PROGRAMMING FA 1 - CSE (AI) 2024 BATCH (MORNING) EXAM

Q1. Alone in Couple

Some people are clubbed together for a party. Everyone in the party is in couple except one. People who are in couple have same numbers. Find out the person who is not in couple.

If there is no alone person, return -1.

Input Format

The first line: An integer representing the total number of people at the party.
The following lines: A sequence of integers (one per line) representing the ID numbers of the people.

Output Format

A single integer representing the ID number of the person who does not have a partner (appears only once) or -1 if every person at the party has a matching partner.

Sample Input

```
7
1
2
3
1
2
4
4
```

Sample Output

```
3
```

Explanation

1, 2, and 4 all have pairs. 3 is the only one left alone.

ANSWER:

Test case 1 :

Input:

```
7
1
2
```

3
1
2
4
4
Output:
3

Test case 2 :

Input:
9
2
2
3
1
1
3
5
4
5
Output:
4

Test case 3 :

Input:
7
1
2
2
3
5
1
3
Output:
5

Test case 4 :

Input:
4
1
2
2
1
Output:
-1

Test case 5 :

Input:
7

```
11
15
15
12
11
16
12
Output:
16
```

Test case 6 :

```
Input:
5
0
0
1
1
2
Output:
2
```

Q2. Find the Intersection of Two Ranges of Numbers

Write a program that finds the intersection of two ranges of numbers. You need to enter 4 numbers in total: the start and end of the first range, and the start and end of the second range. The program should calculate and output the numbers that fall within both ranges.

If no intersection number is possible than print "No intersection".

Input Format

Four integers (one per line):
Line 1: The start of the first range.
Line 2: The end of the first range.
Line 3: The start of the second range.
Line 4: The end of the second range.

Output Format

A sequence of integers, All numbers that exist in both the first and second range, separated by spaces.
If the two ranges do not overlap at all Print "No intersection"
Print space only between numbers, not after the last one

Sample Input

```
1
10
5
15
```

Sample Output

5 6 7 8 9 10

Explanation

The numbers from 5 to 10 are common to both ranges

ANSWER:

Test case 1 :

Input:

1

10

5

15

Output:

5 6 7 8 9 10

Test case 2 :

Input:

-3

2

1

5

Output:

1 2

Test case 3 :

Input:

10

20

15

25

Output:

15 16 17 18 19 20

Test case 4 :

Input:

5

10

1

4

Output:

No intersection

Test case 5 :

Input:

0

5

3
8
Output:
3 4 5

Test case 6 :

Input:
7
12
10
15

Output:
10 11 12

Q3. Game Of Multiply

There are N numbers, all of them are initially equal to **1**. Now, you have to play a game in which there are N rounds.

In the first round, you multiply every number with -1 .

In the second round, you multiply every second number with -2 .

In the third round, you multiply every third number with -3 .

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.

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For the i th round, you multiply every i th number with $-i$.

For the N th round, you multiply the N th number(i.e. the last number) with $-N$.

Return the count of numbers which are less than zero, after N rounds.

Input Format

The first line: An integer representing the number of test cases.

The following lines: For each test case, a single integer N representing N numbers and N rounds

Output Format

For each test case: A single integer representing how many of the N numbers are negative (less than zero) after all N rounds are completed.

Sample Input

2
4

20

Sample Output

2
4

Explanation:

For N = 4, initially the numbers are {1, 1, 1, 1}.
In the 1st round, all the numbers are multiplied with -1, therefore they will become {-1, -1, -1, -1}.
In the 2nd round, every 2nd number is multiplied with -2, therefore they will become {-1, 2, -1, 2}.
In the 3rd round, every 3rd number is multiplied with -3, therefore they will become {-1, 2, 3, 2}.
In the 4th round, the 4th number(i.e. the last number) is multiplied with -4, therefore they will become {-1, 2, 3, -8}.
Hence, after four rounds, 2 numbers are less than zero.

Similarly, for N = 20, 4 numbers will become less than zero, after 20 rounds.

ANSWER:

Test case 1 :

Input:

2
4
20

Output:

2
4

Test case 2 :

Input:

4
0
1
2
3

Output:

0
1
1
1

Test case 3 :

Input:

4
292929299
333333333
900000000
898989880

Output:

17115

18257
30000
29983

Test case 4 :

Input:

3
189
36
678

Output:

13
6
26

Test case 5 :

Input:

4
1000000000
999999999
888888888
777777777

Output:

31622
31622
29814
27888

Test case 6 :

Input:

5
10
10000
41
20
898989

Output:

3
100
6
4
948

Q4. Fibonacci sequence

The Fibonacci sequence is a series of numbers where each number is the sum of the two preceding ones. The sequence starts with 0 and 1.

Your task is to write a function that generates and prints the first n terms of this sequence, starting from 0.

Example Sequence: If n=8 , the sequence is: 0, 1, 1, 2, 3, 5, 8, 13

Input Format

A single integer n , representing the number of terms to be printed.

Output Format

Print the first n terms of the Fibonacci sequence, with each number on a new line.

Constraints

$1 \leq n \leq 70$

Sample Input

4 // n

Sample Output

0
1
1
2

ANSWER:

Test case 1 :

Input:

4

Output:

0

1

1

2

Test case 2 :

Input:

1

Output:

0

Test case 3 :

Input:

2

Output:

0

1

Test case 4 :

Input:

3

Output:

0

1

1

Test case 5 :

Input:

4

Output:

0

1

1

2

Test case 6 :

Input:

70

Output:

0

1

1

2

3

5

8

13

21

34

55

89

144

233

377

610

987

1597

2584

4181

6765

10946

17711

28657

46368

75025

121393
196418
317811
514229
832040
1346269
2178309
3524578
5702887
9227465
14930352
24157817
39088169
63245986
102334155
165580141
267914296
433494437
701408733
1134903170
1836311903
2971215073
4807526976
7778742049
12586269025
20365011074
32951280099
53316291173
86267571272
139583862445
225851433717
365435296162
591286729879
956722026041
1548008755920
2504730781961
4052739537881
6557470319842
10610209857723
17167680177565
27777890035288
44945570212853
72723460248141
117669030460994

Test case 7 :

Input:

5

Output:

0

1

1
2
3

Q5. Sum of Range

Write a function **sumOfRange** that accepts two integer parameters *min* and *max* and returns the sum of all integers from *min* through *max* inclusive.

If *min* is greater than *max*, the function should return 0. Otherwise, it should return the sum of integers in the range min to max.

Input Format:

The first line contains an integer T, the number of test cases.

For each test case, two lines follow:

The first line contains the integer min. The second line contains the integer max.

Output Format:

For each test case, print the sum of integers from min to max inclusive.

If min > max, print 0.

Each result should be printed on a new line with clear identification of the test case or range.

Sample Input:

2
3
7
4
5

Sample Output:

25
9

ANSWER:

Test case 1 :

Input:

2
3
7
4
5

Output:

25
9

Test case 2 :

Input:

6
3
7
0
0
10
9
2
2
-10
-5
-3
10

Output:

25
0
0
2
-45
49

Test case 3 :

Input:

2
4
6
0
5

Output:

15
15

Test case 4 :

Input:

3
15
8
0
10
-5
5

Output:

0
55
0

Test case 5 :

Input:

2

7
7
-2
2
Output:
7
0

Test case 6 :

Input:
2
-10
-1
100
105
Output:
-55
615

Q6. How many numbers are equal to its reverse

Write a program to check how many palindrome numbers are present between a and b. Palindrome Number are those numbers which are equal to their reverse.

For Example 757 is a palindrome number, where 754 is not a Palindrome number.

Input Format

Two integers separated by a space:
The first integer represents the start of the range (inclusive).
The second integer represents the end of the range (inclusive).

Output Format

A single integer that represents total count of numbers within that range that read the same forwards and backwards.

Sample Input

100 111

Sample Output

2

Explanation:

For Input : 100 111
There are only 2 numbers between them that are palindrome, i.e. numbers which are equal to their reverse. Those numbers are 101 & 111

ANSWER:

Test case 1 :

Input:
100 111
Output:
2

Test case 2 :

Input:
110 825
Output:
71

Test case 3 :

Input:
494 584
Output:
9

Test case 4 :

Input:
3478 6594
Output:
31

Test case 5 :

Input:
480 7918
Output:
121

Test case 6 :

Input:
1882 4556
Output:
27

Test case 7 :

Input:
2882 7777
Output:
50

Test case 8 :

Input:
1869 7511
Output:
57

Test case 9 :

Input:
1271 2979
Output:
16

Test case 10 :

Input:
7314 9919
Output:
26

Q7. Given amount, count notes in it

Write a program that accepts an integer amount from the user and determines the **minimum number of currency notes** (2000, 500, 100, 50, 20, 10, 5, 2, 1) required to make that amount using Indian currency denominations.

Available Denominations:

The available denominations are:

2000, 500, 100, 50, 20, 10, 5, 2, 1

Input Format:

A single integer representing the amount.

Output Format:

Print nine space-separated integers.

Each integer represents the number of notes required for the corresponding denomination in the following order:

2000 500 100 50 20 10 5 2 1

Do not print any extra spaces.

Sample Input

2050

Sample Output

1 0 0 1 0 0 0 0 0

Explanation:

1 note of 2000 and 1 note of 50 is required, so print 1 at those places and print 0 on other denominations.

ANSWER:

Test case 1 :

Input:
2050
Output:
1 0 0 1 0 0 0 0 0

Test case 2 :

Input:
4567
Output:
2 1 0 1 0 1 1 1 0

Test case 3 :

Input:
1
Output:
0 0 0 0 0 0 0 0 1

Test case 4 :

Input:
2000
Output:
1 0 0 0 0 0 0 0 0

Test case 5 :

Input:
50
Output:
0 0 0 1 0 0 0 0 0

Test case 6 :

Input:
9876543
Output:
4938 1 0 0 2 0 0 1 1

Q8. Decimal to Binary

Given a decimal number **n**, print its binary equivalent.

Input Format

First line contains a **T** that denotes the number of test cases. This is followed by **T** lines, each containing an integer, **n**.

Constraints

1 <= **T** <= 100
1 <= **n** <= 2³¹ - 1

Output Format

For each test case, print its binary equivalent in a new line.

Sample Input

```
2 // Test Cases
4
11
```

Sample Output

```
100
1011
```

ANSWER:

Test case 1 :

Input:

```
2
4
11
```

Output:

```
100
1011
```

Test case 2 :

Input:

```
10
1
3
2
7
65431
100
20
999999
111111
11100111
```

Output:

```
1
11
10
111
1111111110010111
1100100
10100
11110100001000111111
```

11011001000000111
10101001010111111001111

Test case 3 :

Input:

3
128
257
2147483647

Output:

10000000
100000001
11111111111111111111111111111111

Test case 4 :

Input:

4
292929299
333333333
900000000
898989880

Output:

10001011101011011111100010011
10011110111100100001101010101
110101101001001110100100000000
11010110010101011111100111000

Test case 5 :

Input:

3
5
10
15

Output:

101
1010
1111

Test case 6 :

Input:

7
1
1
2
5
10
15
31

Output:

1
1
10
101
1010
1111
11111

Q9. Leap Year

Write a program to find whether the year is leap or not? A year is considered a leap year if it is perfectly divisible by 4. Years that are perfectly divisible by 100 (for example, 2000, 2100, 2200) are exceptions. Such a year is a leap year only if it is also perfectly divisible by 400.

Input

Each test case will contain a year greater than 1000 AD and less than 10000 AD.

Output

Print "Leap Year" if the given year is a leap year and "Not a Leap Year" otherwise

Sample Input

2017

Sample Output

Not a Leap Year

ANSWER:

Test case 1 :

Input:

2017

Output:

Not a Leap Year

Test case 2 :

Input:

2000

Output:

Leap Year

Test case 3 :

Input:

2017

Output:
Not a Leap Year

Test case 4 :

Input:
2008
Output:
Leap Year

Test case 5 :

Input:
2024
Output:
Leap Year

Test case 6 :

Input:
2100
Output:
Not a Leap Year

Q10. Profit and Loss

If cost and selling price of an item is entered through the keyboard. Write a program to find how much profit/loss is made on that item.

Input Format

Each test case contains 2 values (first is cost price and second is selling price), in two separate lines.

Output Format

Print "Profit" in the first line and profit amount in the second line if profit is made, OR
Print "Loss" in the first line and loss amount in the second line for loss OR
Print "No Profit No Loss" otherwise.

Sample Input:

500
550

Sample Output:

Profit
50

ANSWER:

Test case 1 :

Input:
500
550
Output:
Profit
50

Test case 2 :

Input:
500
500
Output:
No Profit No Loss

Test case 3 :

Input:
450
400
Output:
Loss
50

Test case 4 :

Input:
499
500
Output:
Profit
1

Test case 5 :

Input:
1000
1000
Output:
No Profit No Loss

Test case 6 :

Input:
1000
750
Output:
Loss
250

Write a program to print the first n lines of the following pyramid pattern:

```
1
212
32123
4321234
543212345
```

Input

A single integer n ($1 \leq n$) representing the number of lines of the pyramid.

Output

Print the first n lines of the pyramid

Important Note: Don't print spaces at the end of the row.

Sample Input

3

Sample Output

```
1
212
32123
```

ANSWER:

Test case 1 :

Input:

5

Output:

```
1
212
32123
4321234
543212345
```

Test case 2 :

Input:

4

Output:

1
212
32123
4321234

Test case 3 :

Input:
3
Output:
1
212
32123

Test case 4 :

Input:
8
Output:
1
212
32123
4321234
543212345
65432123456
7654321234567
876543212345678

Test case 5 :

Input:
2
Output:
1
212

Test case 6 :

Input:
7
Output:
1
212
32123
4321234
543212345
65432123456
7654321234567

Write a program to print the n lines of the below pyramid

```
1
2 3
4 5 6
7 8 9 10
```

Input

Each test case will contain a number n where $n \geq 1$.

Output

Print the first n lines of the pyramid

Important Note: Don't print spaces at the end of the row.

Sample Input

```
3
```

Sample Output

```
1
2 3
4 5 6
```

ANSWER:

Test case 1 :

Input:

```
1
```

Output:

```
1
```

Test case 2 :

Input:

```
2
```

Output:

```
1
```

```
2 3
```

Test case 3 :

Input:

3

Output:

1

2 3

4 5 6

Test case 4 :

Input:

4

Output:

1

2 3

4 5 6

7 8 9 10

Test case 5 :

Input:

5

Output:

1

2 3

4 5 6

7 8 9 10

11 12 13 14 15

Test case 6 :

Input:

10

Output:

1

2 3

4 5 6

7 8 9 10

11 12 13 14 15

16 17 18 19 20 21

22 23 24 25 26 27 28

29 30 31 32 33 34 35 36

37 38 39 40 41 42 43 44 45

46 47 48 49 50 51 52 53 54 55